MODULE I

FUNDAMENTALS OF COMPUTER SYSTEMS

BASICS OF OPERATING SYSTEM

- An OS is a program that controls the execution of application programs and **acts as an interface between applications and the computer hardware.**

It can be thought of as having three objectives:

• **Convenience:** An OS makes a computer more convenient to use.

• **Efficiency:** An OS allows the computer system resources to be used in an efficient manner.

• **Ability to evolve:** An OS should be constructed in such a way as to permit the effective development, testing, and introduction of new system functions with-out interfering with service.

FUNCTIONS PERFORMED BY OS

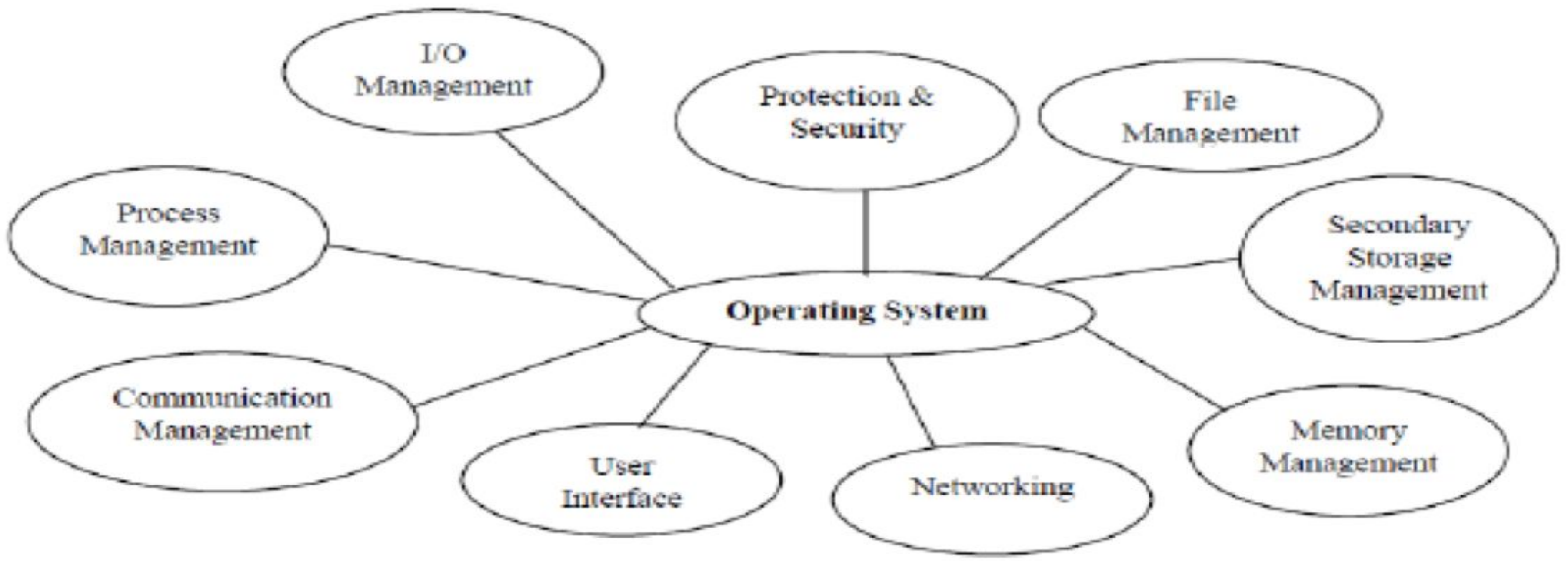


Fig 4: Functions performed by Operating System

PROCESS MANAGEMENT

- The CPU executes a large number of programs.

  A process is a program in execution.

The operating system is responsible for the following activities in connection with processes management:

- The **creation and deletion** of both user and system processes.
- The **suspension and resumption** of processes.
- The provision of mechanisms for **process synchronization.**

MEMORY MANAGEMENT

The operating system is responsible for the following activities in connection with memory management:

- **Keep track of which parts of memory are currently being used** and by whom.
- The OS must **prevent independent processes from interfering** with each other’s memory, both data and instructions.
- Decide **which processes are to be loaded into memory** when memory space becomes available.
- **Allocate and de allocates** memory space as needed.

SECONDARY STORAGE MANAGEMENT

- The main purpose of a computer system is to **execute programs.**
- These programs, together with the data they access, must be in main memory during execution.
- **Since the main memory is too small to permanently accommodate all data and program, the computer system must provide secondary storage to backup main memory.**

The operating system is responsible for the following activities in connection with disk management:

- **Free space management**
- **Storage allocation**

FILE MANAGEMENT

- File management is one of the most visible services of an operating system.
- **Files are mapped, by the operating system, onto physical devices.**
- A file is a collection of related information defined by its creator.

The operating system is responsible for the following activities in connection to the file management:

- **The creation and deletion of files.**
- **The creation and deletion of directory.**
- **The mapping of files onto disk storage.**
- **Backup of files on stable (non-volatile) storage.**
- **Protection and security of the files.**

PROTECTION

- The various processes in an operating system must be protected from each other's activities.
- For that purpose, various mechanisms which can be used to ensure that the files, memory segment, CPU and other resources can be operated on only by those processes and users that have **gained proper authorization from the operating system.**

The operating system is responsible for the following activities in connection to the protection:

- Provides a **mechanism for controlling the access of programs, processes, or users to the resources defined by a computer controls.**
- Improve reliability by **detecting errors at the interfaces** between component subsystems.
- Provide a means of **authentication and authorization.**

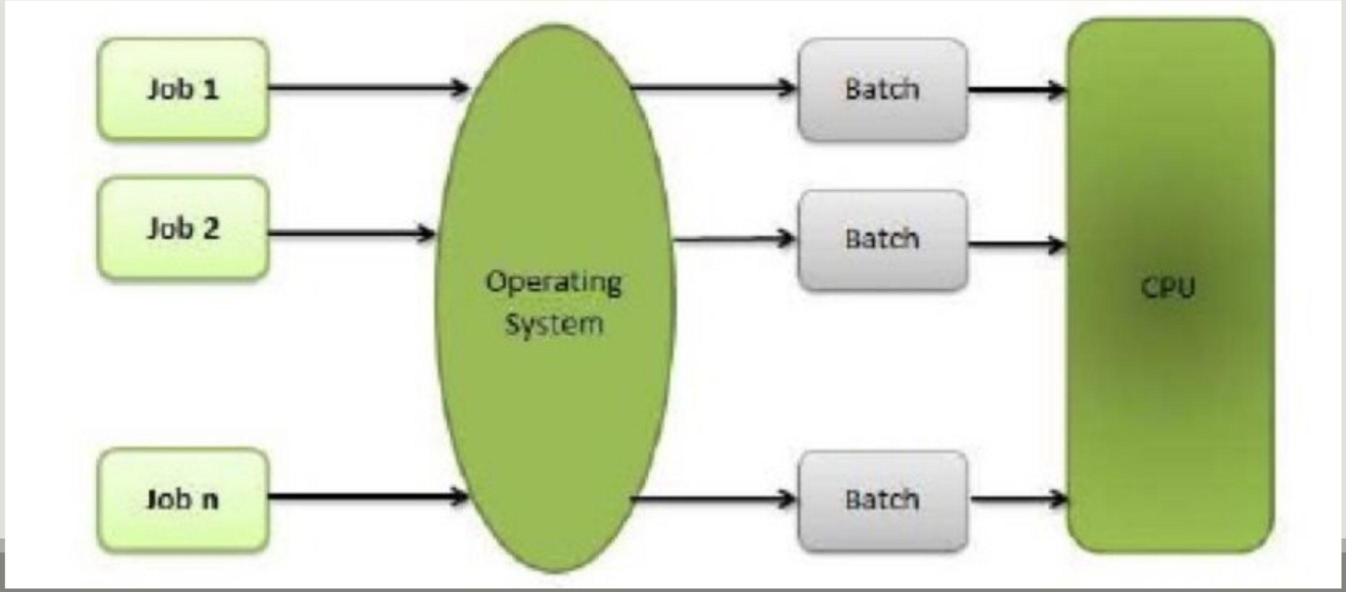
NETWORKING

- With the advancement in technology, **need of distributed system and sharing or accessing information from remote computer has become the order of day.**
- Almost every organization is dependent of Intranet or Internet. So there is a need of networking protocols and their implementation.
  - The operating system is responsible for the following activities in connection to the networking:
- **Setup of a networking connection (Wireless & LAN).**
- **Mechanism for maintaining IP Address.**
- **Managing networking protocol.**
- **Security of network connection.**
- **Firewall management for blocking unauthorized access of computer.**

TYPES OF OPERATING SYSTEMS

Batch Processing

1. “Batch is defined as a **group of job with similar needs and similar resource requirements”.**
2. The operating systems that allow users to create batches and **execute each batch sequentially, processing all jobs of a batch considering them as a single process is called Batch Processing System.**



Advantages of Batch Processing

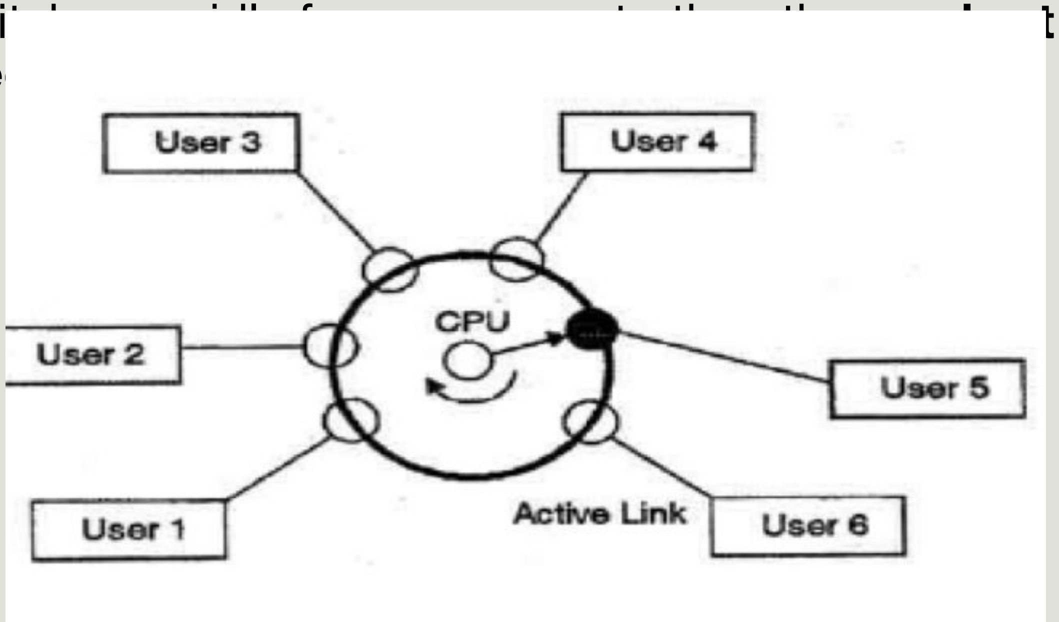
1. Once the data process is started, **the computer can be left running without supervision.**
2. Allows an organization to **increase efficiency** because a large amount of transactions can be combined into a batch rather than processing them each individually.

Disadvantages of Batch Processing

1. With batch processing **there is a time delay before the work is processed and returned**.
2. It is **very difficult to maintain the priority between the batches.**
3. There is **no direct interaction of user with computer.**

Time Sharing Operating System

1. A time sharing system **allows many users to share the computer resources simultaneously.**
2. In other words, time sharing refers to the **allocation of computer resources in time slots to several programs simultaneously.**
3. As the system swi **t time slot is given to each user for their exe**



Advantages of Time Sharing System

1. More than one user can execute their task simultaneously.
2. CPU Idle time is reduced and better utilization of resources.

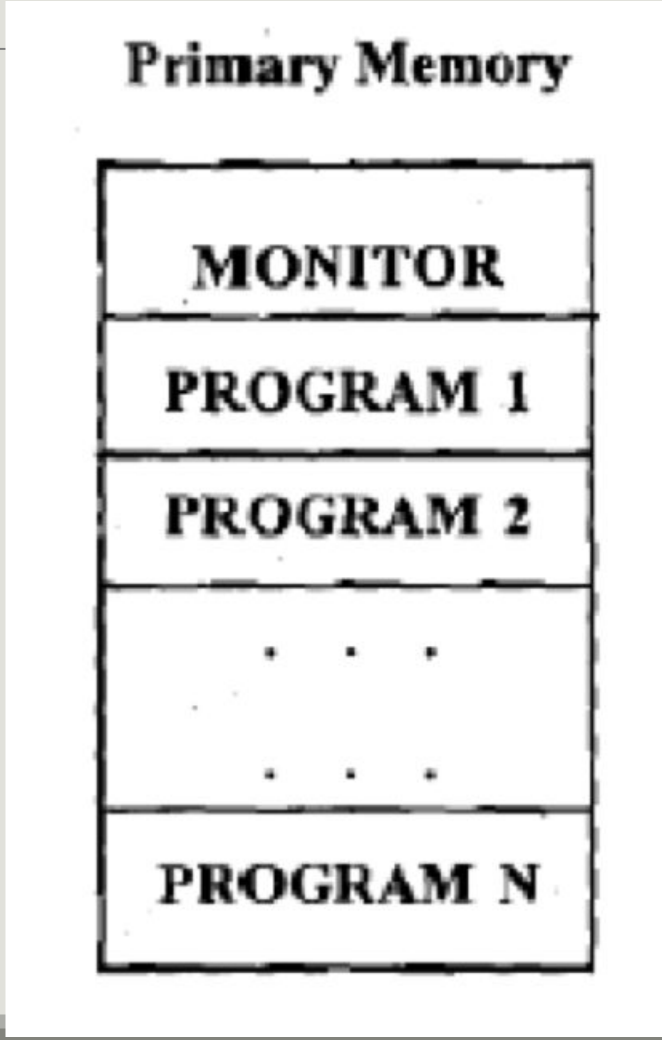
Disadvantages of Time Sharing System

1. Since multiple processes are managed simultaneously, so it requires an adequate management of main memory.

Multi Programmed Operating System

1. A multiprogramming operating system is a system that **allows more than one active user program (or part of user program) to be stored in main memory simultaneously.**

2. The operating system **picks one of the programs and starts executing.**



Advantages of Multi Programming Operating System

1. **Increase CPU utilization and reduce CPU idle time**.

Disadvantages of Multi Programming Operating System

1. It is **fairly sophisticated and more complex** as compared to Uni-programming system.
2. Process can **consume more than available memory. In such case system slows down or it may hang some time.**

Real Time Operating System (RTOS)

1. It is an operating system (OS) intended to serve **real-time application requests.**

2. It must be able to p**rocess data as it comes in, typically without buffering delays.** These operating system handles those application where response time is of extreme importance.

3. These operating systems are used to **control machinery, scientific instruments and industrial systems.**

There are two types of real time systems:

Soft Real Time Systems:

1. If certain deadlines are missed then system continues working with no failures and **it doesn’t cause any severe disaster but its performance will degrade.**
2. It is **less restrictive type** of Real Time System.
3. Example of Soft Real Time system is **MP3 players on mobile**.

Hard Real Time Systems:

1. If any deadlines are missed then then the system will not work and it will **result in severe disasters.**
2. It is **completely restrictive type** of Real Time System.
3. Example of Hard Real Time system **is brakes in auto mobile, rocket launching, flight controls etc**.

Advantages of Real Time Operating System

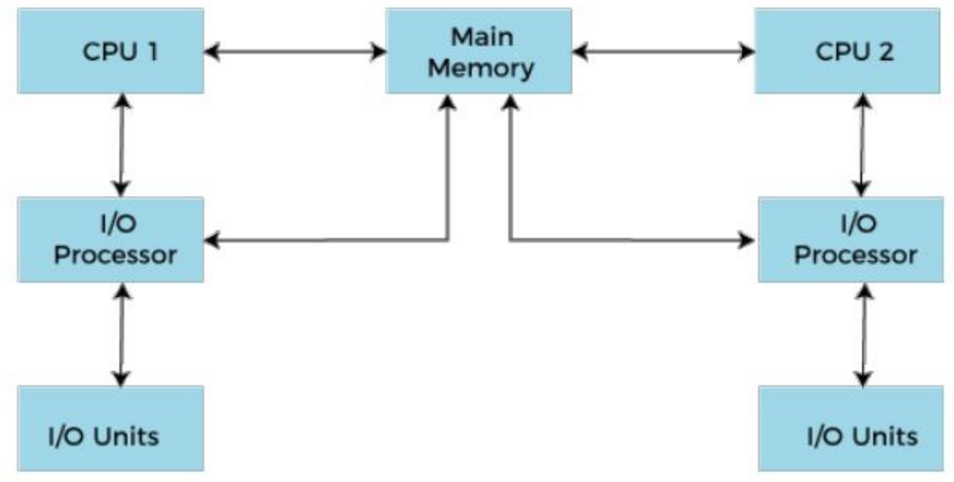
1. **Better task scheduling** as compared to manual process and time deadlines achievement is guaranteed in most of the cases.

Disadvantages of Real Time Operating System

1. If **time dead lines are missed** it may result in **severe disastrous situation.**
2. Complex and need additional kernel, Memory and other resources are required.

Multiprocessing Operating System

1. There are **more than one processors present in the system which can execute more than one process at the same time.**
2. This will increase the throughput of the system.



Working of Multiprocessor System

Advantages of Multi-Processing Operating System

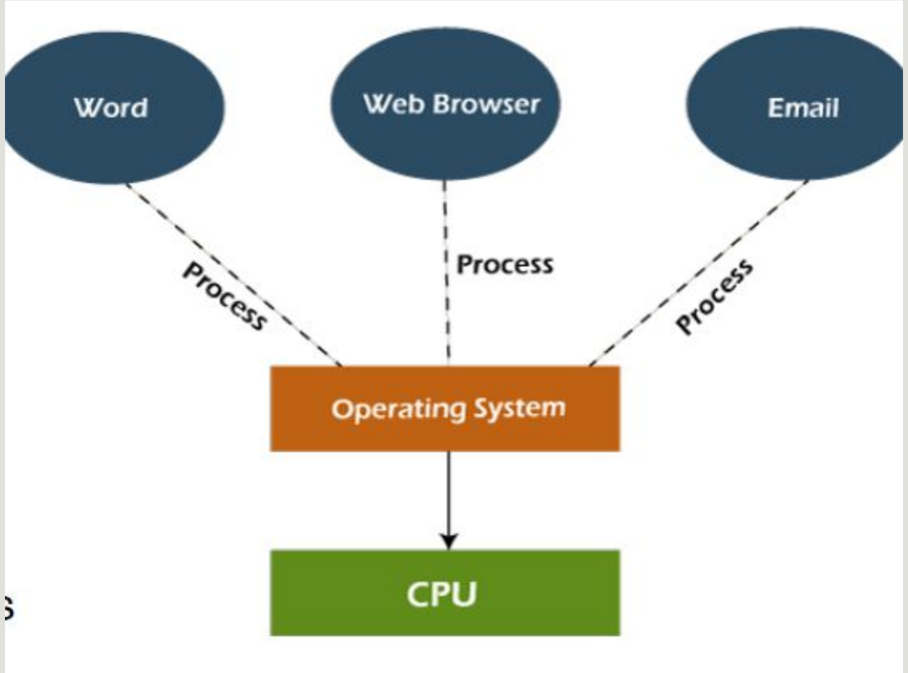
1. Due to multiplicity of processors, multiprocessor systems have better performance than single processor systems.
2. In a properly designed multiprocessor system, if one of the processors breaks down, **the other processor(s) automatically takes over the system workload until repairs are made.** Hence, a complete breakdown of such systems can be avoided.

Disadvantages of Multi-Processing Operating System

1. **Expensive to procure and maintain** so these systems are not suited to daily use.

Multitasking Operating System

1. Multitasking operating systems **allow more than one program to run at a time.**
2. It does this by **dividing system resources amongst these tasks/jobs/ processes and switching between the tasks/jobs/processes while they are executing very fast over and over again.**



Advantages of Multitasking Operating System

1. Multitasking **increases CPU utilization.**
2. **Multiple tasks can be handled at a given time.**

Disadvantages of Multitasking Operating System

1. To perform multitasking the **speed of processor must be very high to daily use.**
2. There are chances that **the computer might hang while handling multiple processes.**

BOOT PROCESS

Booting refers to the **process of loading an operating system image into computer memory and starting up the operating system.**

There are two forms of booting: **Cold Booting and Warm Booting**

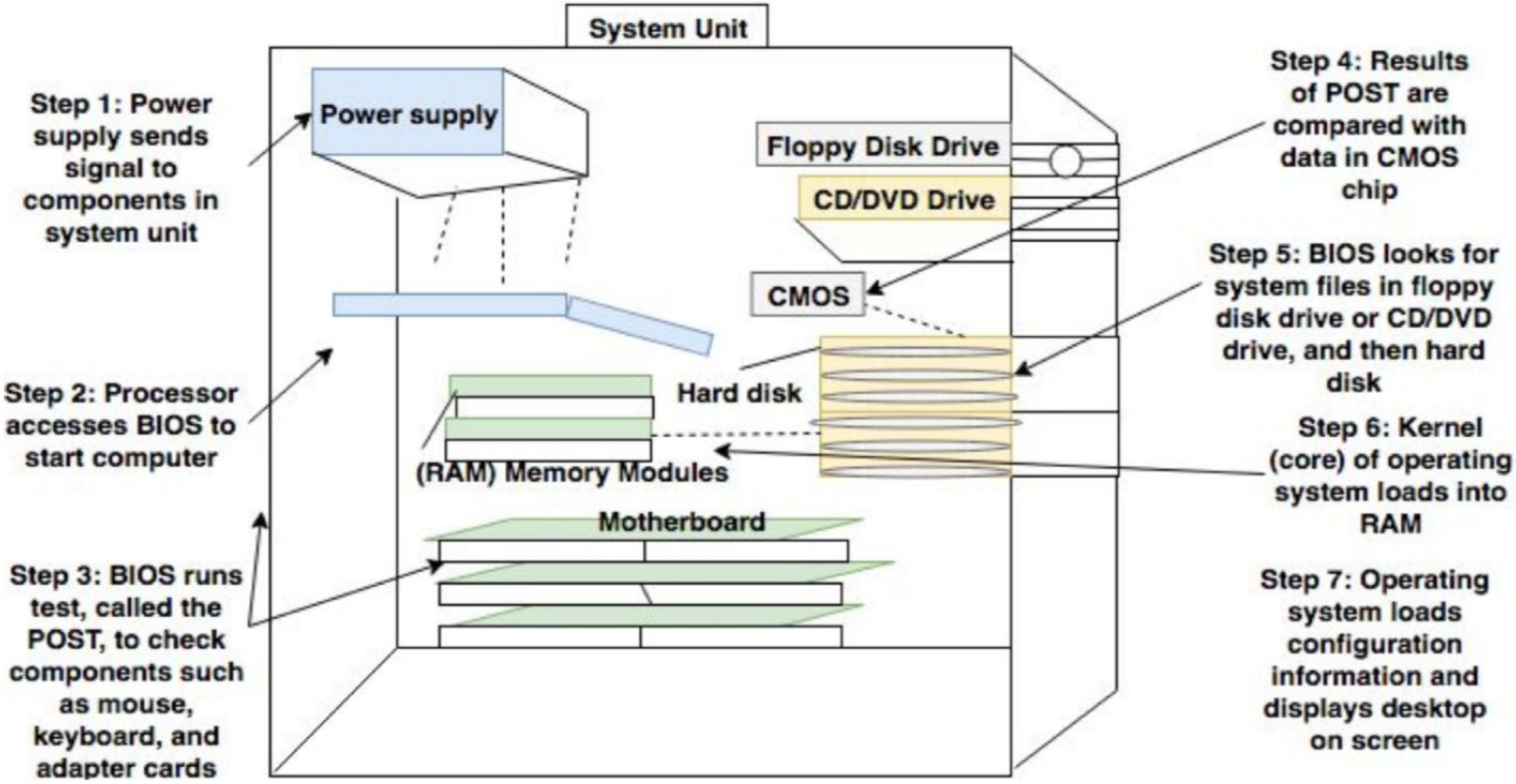
Cold booting(Hard Booting) is the process when we first start the computer.

1. In other words, when the **computer is started from its initial state by pressing the power button it is called cold boot.**
2. **The instructions are read from the ROM and the operating system is loaded in the main memory.**

Warm booting(Soft booting) refers to when we restart the computer.

1. Here, the computer does not start from the initial state.
2. When the **system gets stuck sometimes it is required to restart it while it is ON. Therefore, in this condition the warm boot takes place.**

Boot Process



INITIATING THE BOOT PROCESS

1. When a computer/PC is **booted up or turned on, power is being generated to the system that flows to several internal components.**

2. Tests of all the voltage and current levels are carried out by the power supply.

3. When power is stable and all levels are acceptable, the **power supply sends a Power Good (PWRGD) signal to the processor.**

4. This is **controlled by *BIOS* or the firmware stored on a flash memory chip found in the *motherboard.***

5. The **BIOS settings are stored separately in CMOS memory, which requires a small battery to maintain its integrity.**

POST (POWER-ON SELF-TEST)

1. After the power supply is activated, the PC moves on to the **next stage known as *POST*** which is a small computer program within *BIOS* **used to evaluate hardware and ensure that these physical parts are not experiencing any setbacks.**

2. *POST* can **easily detect problems from all of the hardware components** including the processor, monitor, power supply, hard drive, motherboard, RAM memory modules, etc.

3. **If a single beep is sounded from the PC, then there are no hardware issues present in the system.**

4. However, an **alternative beep sequence indicates that the PC has detected a hardware issue that needs to be resolved before moving on to the next stages of the process.**

MASTER BOOT RECORD

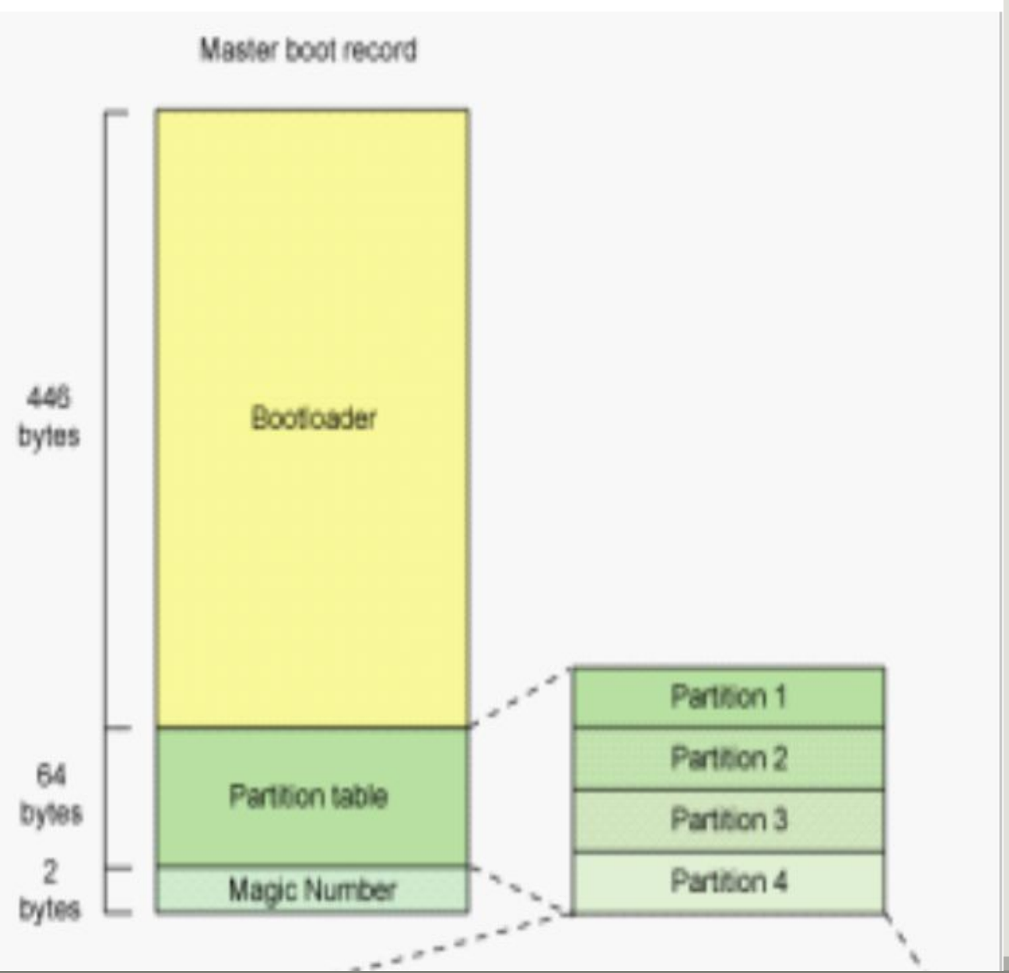
1. The Master Boot Record (MBR) is the **information in the first sector of any hard disk or diskette that identifies how and where an operating system is located so that it can be boot (loaded) into the computer's main memory.**

2. The Master Boot Record is also sometimes called the **"partition sector" or the "master partition table"** because it includes a table that locates each partition that the hard disk has been formatted into.

3. The Master Boot Record (MBR) is traditionally the first 512 bytes, or the first sector (sector 0), of a storage device like a hard disk drive that is bootable.

In the Master Boot Sector of 512 bytes:

1. **Boot Loader take up 446 bytes.**
2. **The second part Disk Partition Table (DPT) takes up 6**
3. **third part is a 2 byte magic number.**



BOOT LOADER

1. BIOS attempts to **access the first available storage address in the first sector of the *boot disk* or hard disk that runs the operating system.**

2. After the *BIOS* software identifies a particular storage address in the *boot disk*, it **runs the *boot loader***.

3. The ***boot loader* or boot manager is generally a small program that places the operating system of a computer into memory or *RAM*.**

4. **The boot code is in the first 446 bytes** of the first 512-byte sector, which is the MBR.

5. It focuses on **loading and starting the boot time tasks and processes found within the operating system**.

TRANSFERRING CONTROL

1. **Once the *boot loader* has successfully placed the OS into memory, the operating system seizes control from the boot process.**

2. The operating system will then finalize any remaining tasks left such as executing pre-configured startup routines.

3. **The moment that there are no more tasks left to be completed, a visual with content will be displayed on the monitor.**